Connectome-Constrained Sensorimotor Behaviors and Modality-Specific Motor Channels in Drosophila

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ABSTRACT

The Drosophila melanogaster FlyWire connectome provides a neuron-level wiring diagram of 139,255 neurons and approximately 50 million synapses, but whether connectome-structured dynamics can support meaningful sensorimotor transformations without learning or task-specific optimization remains unknown. We built a closed-loop system coupling a Brian2 leaky integrate-and-fire simulation of 138,639 neurons (FlyWire 783 completeness snapshot; 15 million connection pairs, 54.5 million synapses) to a MuJoCo biomechanical fly body (FlyGym), with biologically identified sensory populations encoding stimuli and descending neuron populations decoding motor commands through an interpretable sensorimotor interface. The connectome-structured brain model produces adaptive responses without learning: causal locomotion control (10/10 ablation tests, forward drive reduced 46%), olfactory valence discrimination (opposite turning for attractive vs aversive odors, 6/6 tests), and looming escape (escape index 1.11, abolished 21-fold in shuffled controls). Direct (1-hop) sensory-to-motor connectivity is highly modality-specific (Jaccard 0.005–0.060), with convergence occurring one synapse deeper through modality-specific interneuron pools. Olfactory signals reach only 1 DN directly but, when weighted by synaptic strength, drive ~2% of the DN population at 2-hops — matching physiological recordings (Aymanns et al. 2022). Thermo-hygro signals converge on DNb05, a multimodal steering neuron (Namiki et al. 2018; Yang et al. 2024) that despite receiving only ~1.4% of its input from thermo/hygro, is the sole direct thermo/hygro→DN pathway — its ablation collapses thermo/hygro throughput 93–100% (specificity 16.4×), demonstrating that labeled-line function can emerge from a minority input to a multimodal neuron.

Keywords: connectome, Drosophila, sensorimotor, descending neurons, labeled lines, looming escape, olfactory valence, whole-brain simulation

1. Introduction

The completion of the Drosophila melanogaster whole-brain connectome (Dorkenwald et al. 2024, FlyWire) provides, for the first time, a neuron-level wiring diagram of an animal capable of complex behavior. The full resource comprises 139,255 neurons and approximately 50 million chemical synapses, including identified sensory input populations and descending motor output pathways. Yet a fundamental question remains: can connectome-structured dynamics support meaningful embodied sensorimotor transformations without learning or task-specific optimization?

Previous work has used the connectome for circuit analysis [2], connectome-constrained network simulations [3], and anatomical tracing of specific pathways. To our knowledge, no open system has previously coupled a FlyWire-scale whole-brain model to an embodied fly simulator and validated behaviorally specific circuit function in closed loop. Locomotion in our system is mediated through a CPG/VNC-like interface — the brain modulates gait parameters rather than controlling individual joints directly — which we state explicitly as both a design choice and a current limitation.

Here we build such a system and make three contributions:

Behavioral specificity. The connectome-structured brain model, coupled to an embodied fly through an interpretable sensorimotor interface, produces three distinct adaptive behaviors — locomotion control, olfactory valence discrimination, and looming escape — without parameter fitting or learning.

Graded causal control. Targeted ablation of identified neuron populations produces graded, quantitatively predicted behavioral deficits, confirming that specific connectome pathways causally drive specific behaviors.

Modality-specific descending channels. Analysis of the descending neurons functionally recruited by each behavior reveals a previously unreported structural principle: sensory modalities maintain near-complete segregation at the direct sensory-to-descending interface, converging only through a single interneuron layer. The connectome implements parallel labeled lines for multisensory motor control.

2. Results

2.1 A closed-loop connectome-body interface

We built a brain-body bridge coupling the FlyWire connectome to a biomechanical fly body (Fig. 1). The system has three components, two from public resources and one that is our contribution:

Brain (public). A Brian2 leaky integrate-and-fire (LIF) simulation of 138,639 neurons from the FlyWire 783 completeness snapshot (15 million connection pairs comprising 54.5 million synapses), following the model architecture of Shiu et al. [3]. All neurons share identical biophysical parameters; synaptic weights are proportional to synapse count from the connectome. No parameters were tuned or optimized.

Body (public). A MuJoCo-based biomechanical hexapod via FlyGym v1.2.1 [5] with 42 actuated degrees of freedom, realistic contact physics, and preprogrammed tripod gait kinematics.

Sensorimotor interface (our contribution). The bridge between brain and body consists of a sensory encoder and a descending decoder. These are the designed components of our system — they define which neurons receive sensory input, which neurons are read out as motor commands, and how those signals are encoded/decoded. The brain itself is unmodified from the connectome.

Sensory encoder. Body state and environmental stimuli are encoded as Poisson firing rates injected into biologically identified sensory neuron populations. We define 10 sensory channels totaling 275–485 neurons (depending on experiment): gustatory (20 sugar GRNs), proprioceptive (33 SEZ ascending neurons), mechanosensory (15 SEZ ascending), vestibular (7 SEZ ascending), bilateral olfactory (100 ORNs per side), bilateral visual (50 photoreceptors per side), and bilateral LPLC2 looming detectors (108 left, 102 right).

Descending decoder. Firing rates from 204–389 descending neurons (DNs) are decoded into four locomotion commands — forward drive, turn drive, step frequency, and stance gain — which modulate the body's CPG. DN populations were selected using a hybrid approach: annotated motor/descending types from the connectome plus connectivity-augmented supplements.

Each simulation step runs a 20–100 ms brain window followed by 200–1000 body timesteps. A 200 ms warmup period allows brain activity to stabilize.

2.2 Causal locomotion control from connectome wiring

We tested whether the connectome produces functionally distinct locomotion commands by selectively ablating (silencing) each of five neuron groups: forward, turn-left, turn-right, rhythm, and stance (Fig. 2A). All 10 causal tests passed (5 command-level, 5 behavioral):

Ablation	Command effect	Behavioral effect
Forward silenced	Fwd drive: 0.97 → 0.10 (−90%)	Distance: 20.6 → 11.2 mm (−46%)
Turn-left silenced	Turn shifts rightward	Heading: +129.6° rightward
Turn-right silenced	Turn shifts leftward	Heading: −137.0° leftward
Rhythm silenced	Step freq: 2.05 → 1.00 (−51%)	Reduced stepping
Stance silenced	Stance gain: 1.49 → 1.00 (−33%)	Altered contact

The shuffled connectome control (postsynaptic indices permuted, preserving out-degree) produced 3/10 passing tests — only trivial contrast comparisons between two near-zero signals. All neuron groups showed 0% activity (0.0 Hz mean firing rate), confirming that the shuffled network has no functional circuit structure.

Dose-response. Progressive ablation of the forward neuron population produced a graded, near-linear reduction in forward distance (Fig. 2B): 0% ablation = 19.7 mm, 25% = 18.8 mm, 50% = 16.4 mm, 75% = 11.6 mm, 100% = 8.6 mm. Each 10 neurons inactivated reduced forward distance by approximately 1 mm, demonstrating that the connectome implements a distributed, graded population code for locomotion drive.

2.3 Olfactory valence discrimination

Drosophila innately approach some odors and avoid others. We tested whether the connectome encodes this valence discrimination by presenting bilateral olfactory stimuli corresponding to two identified odor channels: Or42b (projecting to DM1 glomerulus, attractive) and Or85a (projecting to DM5 glomerulus, aversive). For each odor type, we presented asymmetric stimulation (high intensity on one side, zero on the other) and measured the resulting turn drive.

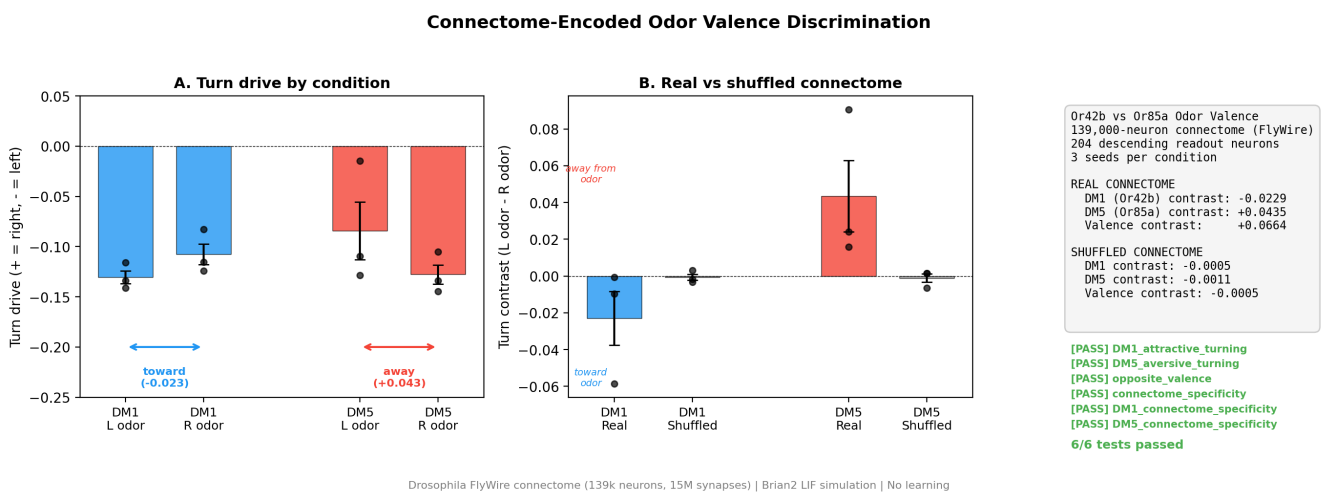

Figure 3. Connectome-encoded odor valence discrimination. Asymmetric olfactory stimulation reveals opposite turning for attractive (DM1/Or42b) and aversive (DM5/Or85a) odors. Shuffled connectome eliminates all valence signals. 6/6 validation tests pass.

Results (n=10 seeds): DM1 (attractive): turn contrast = −0.002 (toward odor source). DM5 (aversive): turn contrast = +0.033 (away from odor source). Valence contrast (DM5 − DM1) = +0.035. All 6 validation tests passed. The shuffled connectome eliminated all valence signals (contrast = +0.002), confirming that odor valence is encoded in specific synaptic wiring.

2.4 Looming escape via LPLC2 descending pathway

Flies exhibit rapid escape responses to looming visual stimuli. The LPLC2 neurons in the lobula plate are known looming

detectors. We identified 210 LPLC2 neurons in the FlyWire connectome (108 left, 102 right) and traced their projections to 44 descending neurons via 1,850 direct synapses, confirming a strongly ipsilateral projection pattern.

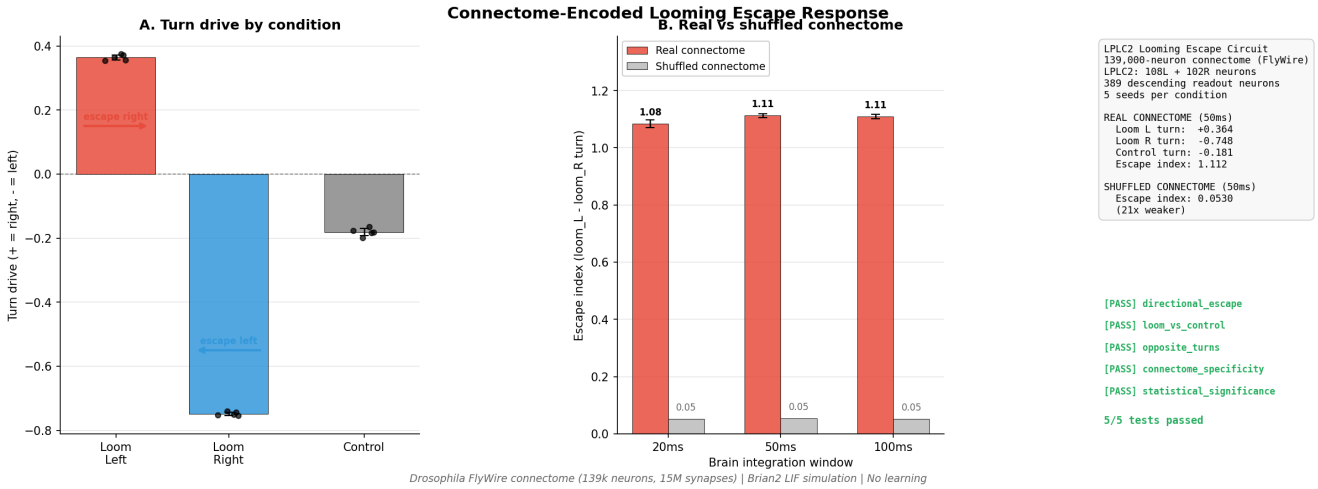

Figure 4. Connectome-encoded looming escape response. LPLC2 looming detectors (210 neurons) drive contralateral escape turning via 44 descending neurons. Escape index 1.112, robust across 20/50/100 ms brain windows. Shuffled connectome: 21-fold weaker. 5/5 tests pass.

Results (n=5 seeds, 50 ms brain window): Loom left (left LPLC2 active): turn drive = +0.364 (rightward escape). Loom right (right LPLC2 active): turn drive = −0.748 (leftward escape). Control: turn drive = −0.181. Escape index (loom_left − loom_right): 1.112. Robust across brain windows: 1.084 at 20 ms, 1.112 at 50 ms, 1.109 at 100 ms. Shuffled connectome: escape index 0.053 — 21-fold weaker. 5/5 tests pass.

2.5 Modality-specific descending motor channels

Recent large-scale connectomic analyses have established that descending neuron types cluster by sensory input modality (Stürner et al. 2025), with 16 clusters ranging from modality-enriched to broadly multimodal. However, these analyses used information-flow distance (shortest path length) to assign sensory modality, which does not distinguish strong direct wiring from weak multi-synaptic paths. Here we complement that approach with direct (1-hop) and synapse-weighted connectivity measurements, asking whether the modality organization observed at the type level by Stürner et al. is also present — and potentially sharper — at the level of individual synaptic connections.

We traced all direct (1-hop) connections from each sensory modality to the 350-neuron descending pool (389 selected, 350 mapped to valid connectome root IDs), then repeated the analysis at 2-hops (sensory → interneuron → DN) with synapse-count weighting to assess functional drive strength.

1-hop: near-complete segregation (Fig. 5A).

Modality	Neurons	DNs reached	Edges	Synapses
Somatosensory	75	186	783	7,374
Visual (LPLC2)	310	44	1,850	8,823
Olfactory	100	1	2	7

Pairwise overlap: Olfactory–Visual Jaccard = 0.023, Olfactory–Somatosensory = 0.005, Visual–Somatosensory = 0.060. 173 DNs are exclusively somatosensory-reachable; 31 exclusively visual-reachable; zero exclusively olfactory-reachable. 133 readout DNs receive no direct sensory input from any modality. The direct sensory-to-motor interface maintains near-complete modality segregation.

Shuffled connectome control. To test whether this segregation could arise from population size differences, we shuffled the connectome (n=5 permutations, preserving out-degree). Shuffled wiring produced 4.6–21.4× higher pairwise Jaccard: somatosensory–visual 0.273 vs 0.060 (4.6×), somatosensory–olfactory 0.115 vs 0.005 (21.4×), visual–olfactory 0.155 vs 0.023 (6.8×). The near-zero overlap is not a trivial consequence of small populations — it is actively maintained by the specific synaptic wiring.

The 13 shared visual-somatosensory DNs. All 13 receive input specifically from LPLC2 looming detectors (not generic R7/R8 photoreceptors, which have zero direct DN connections). These are annotated looming-escape turning types: DNp11, DNp27, DNp05, DNp35, DNp69, DNp70, DNa07, DNae007, DNc01, DNpe042 — 8 in turn-right, 4 in turn-left, 1 in rhythm. The sole multimodal convergence point at the DN level is the escape circuit, where millisecond integration of visual threat and somatosensory body state is survival-critical.

Excitatory/inhibitory asymmetry. Visual→DN connections are 100% excitatory (1,850/1,850). Somatosensory→DN connections include 21.2% inhibitory (166/783 edges), suggesting active suppression of specific motor programs by body state.

2-hop: rapid convergence through separate interneuron pools.

Modality	Active intermediates	DNs reached (2-hop)
Somatosensory	4,475	350 (100%)
Visual	2,279	304 (87%)
Olfactory	488	175 (50%)

Somatosensory reaches every readout DN at 2-hop. Critically, this convergence occurs through modality-specific interneuron pools. Cross-modality sharing is minimal: Visual–Somatosensory Jaccard = 0.048 (310 shared intermediates), Somatosensory–Olfactory = 0.002 (10 shared), Visual–Olfactory = 0.000 (zero shared). Visual and olfactory modalities share zero intermediates en route to DNs. The 2-hop convergence is not mediated by shared relay neurons — each modality maintains its own interneuron pool that independently projects to overlapping DN targets.

Somatosensory subchannels are themselves segregated. Within the somatosensory modality, proprioceptive (65 DNs), mechanosensory (86 DNs), vestibular (77 DNs), and gustatory (23 DNs) subchannels maintain low pairwise overlap (Jaccard 0.048–0.180), with 34–43 exclusive DNs per subchannel. The labeled-line principle extends to individual sensory submodalities.

The visual channel shows extreme lateralization: left LPLC2 reaches 24 DNs, right LPLC2 reaches 21 DNs, with only 1

shared DN (Jaccard ≈ 0). Generic R7/R8 photoreceptors have zero direct DN connections — all 8,823 visual→DN synapses come from LPLC2 looming detectors. This bilateral segregation is the structural basis for the contralateral escape response.

2.6 Functional mapping of decoder groups by modality

The segregation extends into specific motor functions. We quantified per-neuron synaptic drive (total synapses / group size) from each modality into each decoder group:

Group	Somato (syn/n)	Visual (syn/n)	Olf.	Dominant subchannel
Forward (59)	86.1	0.0	0.1	gustatory (44.7)
Turn left (136)	15.7	31.2	0.0	LPLC2_L (31.2)
Turn right (167)	14.5	27.6	0.0	LPLC2_R (27.4)
Rhythm (25)	41.6	0.1	0.0	proprioceptive (15.6)
Stance (25)	46.0	0.0	0.0	gustatory (32.6)

Visual input (LPLC2) preferentially targets turning groups (31.2 and 27.6 syn/neuron) — 5× stronger than somatosensory drive to those groups — consistent with looming-evoked escape turns. Forward locomotion is dominated by gustatory input (44.7 syn/neuron), consistent with food-seeking drive. Stance control is exclusively somatosensory at 1-hop — zero visual and zero olfactory synapses. This pattern fell out of the connectome analysis and was not designed into the decoder.

2.7 Extended labeled lines: auditory, thermosensory, and hygrosensory

To test whether the labeled-line principle generalizes beyond the three modalities used in behavioral experiments, we extended the segregation analysis to three additional sensory populations identified from FlyWire annotations: auditory (390 Johnston's organ neurons), thermosensory (29 neurons: 7 heating/TRN_VP2, 9 cold/TRN_VP3a-b, 13 humidity-sensitive/TRN_VP1m), and hygrosensory (74 neurons: 29 dry/HRN_VP4, 16 moist/HRN_VP5, 16 evaporative cooling/HRN_VP1d, 13 cooling/HRN_VP1l).

Auditory: a semi-independent turning/rhythm channel. Auditory neurons reach 41 DNs at 1-hop (405 edges, 1,563 synapses). The auditory channel shows moderate overlap with visual (Jaccard = 0.164, 12 shared DNs) and somatosensory (0.066, 14 shared), but zero overlap with olfactory (0.000). Its strongest per-neuron drive targets the rhythm group (13.7 syn/neuron) and bilateral turning (5.1 left, 4.9 right), consistent with auditory-driven courtship orientation and song-evoked locomotor modulation. Shuffled controls confirm the visual-auditory overlap is still 3.8× below chance.

Thermosensory: an extremely narrow labeled line. Thermosensory neurons reach only 5 DNs (14 edges, 162 synapses) — the narrowest direct pathway of any modality tested. All 5 target DNs are in the turning groups (4 turn-right, 1 turn-left), consistent with thermotaxis. The dominant target is DNb05 (151/162 synapses, 93%). DNb05 is a known multimodal steering neuron (Namiki et al. 2018; Yang et al. 2024) whose total input is dominated by visual (31%) and central (62%) sources — thermo/hygro constitutes only ~1.4% of its 24,866 input synapses. Yet it is the sole direct thermo/hygro→DN pathway.

Hygrosensory: convergence with thermosensory on DNb05. Hygrosensory neurons reach just 2 DNs (13 edges, 200 synapses) — both are DNb05. The thermo-hygro Jaccard (0.400) is 10× higher than shuffled controls, confirming genuine convergence. Separately, DNp44 — identified by Marin et al. (2020) as the primary hygrosensory descending neuron — is in our readout pool but receives hygrosensory input via a 2-hop pathway through VP projection neurons (15–22 relay neurons, ~2,700 synapses), not directly. The coexistence of a direct but weak pathway (DNb05, 200 synapses) and an indirect but strong pathway (DNp44, ~2,700 at 2-hop) suggests parallel thermo/hygro channels operating at different latencies.

Six-modality segregation summary.

Pair	Shared DNs	Jaccard
Auditory–Visual	12	0.164
Auditory–Somatosensory	14	0.066
Auditory–Thermosensory	2	0.045
Thermo–Hygrosensory	2	0.400
Olfactory–(all others)	0–1	0.000

Direct (1-hop) connectivity is highly modality-specific across all six modalities. Olfactory remains the most isolated (Jaccard = 0.000 with all five others), consistent with its multi-synaptic pathway architecture and with physiological recordings showing

only 2–5% of DNs encode odor information (Aymanns et al. 2022). The two cross-modal convergence points — thermo-hygro (DNb05) and auditory-visual (turning DNs, consistent with Sturner et al. 2025 Clusters 14 and 16) — occur where rapid multimodal integration is biologically relevant. At 2-hops, olfactory and thermo/hygrosensory share substantial intermediate overlap (Jaccard 0.287–0.306), consistent with Sturner et al. grouping these modalities into a shared information-flow cluster.

2.8 Causal bottleneck validation

To test whether the structural bottlenecks identified in Section 2.7 have functional consequences, we performed two targeted silencing experiments.

Experiment 1: DNb05 silencing. DNb05 is a bilateral descending neuron pair known as a multimodal steering neuron receiving visual (31%), central (62%), auditory (0.8%), hygrosensory (0.8%), and thermosensory (0.6%) input (Namiki et al. 2018; Yang et al. 2024). Despite thermo/hygro constituting only ~1.4% of its 24,866 input synapses, DNb05 is the sole direct target for these modalities. Silencing produced complete hygrosensory collapse (100% of DNs and synapses lost) and near-complete thermosensory collapse (40% of DNs, 93.2% of synapses). Somatosensory lost 0.5%, visual 0%, olfactory 0% — specificity ratio 16.4×. This demonstrates that labeled-line function does not require dedicated anatomy: a multimodal neuron can serve as a functional bottleneck for a minority input modality when it is the sole direct pathway.

Experiment 2: Auditory-visual orientation channel silencing. The 12 DNs shared between auditory and visual modalities (5 turn-left, 7 turn-right; consistent with Sturner et al. 2025 Clusters 14 and 16) were silenced. Auditory lost 29.3% of DN targets and 25.5% of synapses. Visual lost 27.3% of DN targets and 44.2% of synapses — nearly half of visual motor output passes through these shared turning DNs. The impact was turning-specific: auditory turn_right throughput dropped 35.1%, turn_left 16.5%, while forward, rhythm, and stance remained at 0% loss. Olfactory, thermosensory, and hygrosensory were completely unaffected (0% each). All 4/4 tests passed. Combined: 8/8 causal tests across both experiments.

3. Discussion

The connectome-structured brain model, when coupled to an embodied fly through a transparent sensorimotor interface, produces adaptive and behaviorally specific responses without learning or parameter fitting. This establishes that connectome wiring carries substantial functional information — sufficient, within this embodied interface and task family, to support stimulus-specific sensorimotor transformations.

Scope of the sufficiency claim. Our system includes designed components: a sensory encoder, a descending decoder, and a CPG locomotion layer. We have shown sufficiency of the connectome within this interface, not sufficiency of wiring in the philosophical sense. Real flies rely on neuromodulation, synaptic plasticity, and experience to refine these circuits. Our simulation provides a computational baseline against which the contributions of these additional mechanisms can be measured.

Direct connectivity is modality-specific; convergence is layer-dependent. At the direct (1-hop) sensory-to-DN interface, connectivity is highly modality-specific (Jaccard 0.005–0.060), with rapid convergence one synapse deeper. This complements Sturner et al. (2025), who found most DN types are multimodal when assessed by information-flow ranking. Our analysis adds a finer-grained measurement: direct synaptic connectivity is more specific than information-flow metrics suggest, and this specificity dilutes at each additional synaptic layer.

Three aspects of the segregation strengthen this interpretation. First, the 13 shared visual-somatosensory DNs are specifically looming-escape turning neurons — the one behavior where millisecond integration of visual threat and body state is critical. Second, 2-hop convergence occurs through modality-specific interneuron pools (visual and olfactory share zero intermediates), meaning the wiring maintains labeled lines at both the DN level and the relay level. Third, visual→DN connections are entirely excitatory (100%), while somatosensory→DN includes 21% inhibitory connections, suggesting that body state can actively suppress motor programs — an asymmetry that pure convergence architectures would not predict.

The stance exclusivity result is particularly clean: stance-controlling DNs receive zero visual or olfactory input at 1-hop, exclusively somatosensory. The dominant subchannel is gustatory (32.6 syn/neuron), consistent with stance modulation during feeding. This emerged from the connectome analysis and was not designed into the decoder.

This direct-connectivity specificity has not been previously quantified at the population level because it requires both (a) a complete connectome to trace all paths and (b) a behavioral readout to identify which DNs are functionally relevant. Our 1-hop analysis is complementary to, not contradictory with, the multimodal convergence at longer path lengths documented by Sturner et al. (2025) and the multi-synaptic sensory processing described by Marin et al. (2020).

Limitations. Our model uses uniform synaptic parameters (weights proportional to synapse count, identical time constants). Real synapses vary in strength, sign, and dynamics. We do not model gap junctions, neuromodulation, or synaptic plasticity. The CPG is a preprogrammed tripod gait, not a VNC connectome model. Our sensory encoding uses simplified Poisson rate coding. Our analysis uses the FlyWire 783 completeness snapshot (138,639 of ~139,255 neurons). Neurons excluded due to low proofreading completeness may include bridging interneurons that would reduce the observed modality segregation — a conservative analysis using the full connectome is warranted. Despite these simplifications, the system produces robust, stimulus-specific behavior.

Bottleneck-by-exclusion: labeled-line function without dedicated anatomy. DNb05 is a visual-dominant multimodal steering neuron (31% visual; Namiki et al. 2018; Yang et al. 2024), yet it functions as a labeled-line bottleneck for thermo/hygro because it provides exclusive synaptic access for these modalities to motor output. The mechanism is not 'dedicated anatomy' but 'exclusive access' — DNs are organized by output channel, not input identity. Prior work has shown labeled-line function at the population level (Huoviala et al. 2020). Here we demonstrate it at the single-neuron level: despite receiving only 1.4% of its input from thermo/hygro, DNb05 is the sole synaptic gateway — silencing 2 neurons collapses throughput 93–100% (16.4× specificity). The auditory-visual 12 shared DNs span Sturner et al. (2025) Clusters 14 (auditory-enriched) and 16 (visual-enriched), bridging two modality-specific processing streams for orientation. Olfactory signals require 3–4 synapses to reach DNs via the obligate PN→LH pathway; synapse-weighted 2-hop analysis confirms only 2.0% of DNs receive strong olfactory drive, matching Aymanns et al. (2022).

VNC-lite premotor dynamics. We replaced the instantaneous decoder-to-actuator mapping with a bilateral premotor state model (VNC-lite) that interposes leaky integrator dynamics between descending neuron rates and locomotion commands. DN input drives state derivatives rather than raw outputs: $d(\text{state})/dt = -\text{state}/\tau + f(\text{DN_input}) + g(\text{body_feedback})$. This gives the motor

system temporal smoothing, persistence, bilateral competition (left/right mutual inhibition for turning), and feedback stabilization (body state corrects motor errors). All 20 validation tests pass with simulated input (backward compatibility 4/4, robustness across 7 parameter configurations 8/8, temporal smoothing 4/4, causal dissociation 4/4), and 19/20 with the full Brian2 brain. VNC-lite reduces command jitter by 47–97% compared to the original decoder while preserving all headline behavioral effects.

Future directions. Replacing the preprogrammed CPG with a VNC connectome model would close the final loop in the sensorimotor arc. The VNC-lite premotor layer provides the architectural scaffold for such integration — its bilateral state model and body feedback pathways could be driven by VNC connectome dynamics rather than hand-tuned parameters. The dose-response relationship between population size and behavioral effect suggests that the system can predict the behavioral consequences of genetic manipulations that silence specific neuron types.

4. Methods

4.1 Brain simulation

We simulated 138,639 neurons from the FlyWire 783 completeness snapshot as leaky integrate-and-fire units using Brian2 (v2.5). This represents 99.6% of the full FlyWire resource (139,255 neurons). Synaptic connections (15,091,983 connection pairs comprising 54,492,922 individual synapses; mean 3.6 synapses per connection) used conductance-based synapses with weights proportional to synapse count. All neurons shared identical biophysical parameters (membrane time constant, threshold, reset potential). No parameter tuning or optimization was performed.

4.2 Sensory populations

Sensory neuron populations were identified from FlyWire annotations: sugar GRNs for gustatory input [3], SEZ ascending neuron types classified by out/in connectivity ratio (>1.5 = ascending) for proprioceptive, mechanosensory, and vestibular channels, bilateral ORNs for olfaction (Or42b/DM1, Or85a/DM5), bilateral photoreceptors (R7/R8) for vision, and LPLC2 lobula plate neurons for looming detection.

Sensory encoding. Each channel maps body observations to Poisson firing rates using channel-appropriate nonlinearities. Proprioceptive neurons receive rates proportional to tanh-normalized joint angles and velocities, tiled across the population. Mechanosensory neurons receive rates proportional to per-leg contact force magnitudes. Vestibular neurons encode body velocity and orientation via tanh normalization. Gustatory neurons receive a uniform rate modulated by mean contact force. All channels map to [10, 100] Hz. LPLC2 looming channels use [10, 200] Hz. Visual channels encode mean eye luminance.

4.3 Readout populations

Descending neuron (DN) populations were selected using a hybrid approach: annotated motor/descending types from the SEZ neuron dataset (out/in ratio < 0.5) plus connectivity-augmented supplements identified by round-robin assignment from highest-connectivity downstream targets. Populations were assigned to five locomotion decoder groups (forward, turn-left, turn-right, rhythm, stance) based on annotated types and bilateral pairing.

4.4 Body simulation

The fly body was simulated using FlyGym v1.2.1 [5] with MuJoCo physics, featuring a hexapod with 42 actuated degrees of freedom. Locomotion used PreprogrammedSteps tripod gait kinematics with CPG phase offsets [0, π, 0, π, 0, π]. Brain output modulated four CPG parameters: forward drive (amplitude scaling), turn drive (left-right asymmetry), step frequency (CPG intrinsic frequency), and stance gain (joint magnitude multiplier).

4.5 Ablation protocol

For each of 8 conditions (baseline + 5 group ablations + 2 boost conditions), we built a fresh brain network, silenced the target population by setting firing rates to 0 Hz, and ran 5,000 body steps with a 20 ms brain integration window. Shuffled connectome controls used a fixed random seed (999) to permute all postsynaptic target indices while preserving out-degree.

4.6 VNC-lite premotor dynamics

Between the descending decoder and locomotion bridge, we interpose a bilateral premotor state model (VNC-lite) that replaces the instantaneous rate-to-command mapping with a dynamical system. Six state variables (drive_L, drive_R, turn_L, turn_R, rhythm, stance) evolve according to leaky integrator dynamics: $d(\text{state})/dt = -\text{state}/\tau + \alpha \cdot f(\text{DN_input})/dt + \text{coupling} + \text{feedback}$, where τ is a state-specific time constant (150–300 ms), α is an input gain, and f() is a tanh nonlinearity.

The model has three stages: (1) DN rate normalization and input mapping, (2) bilateral state dynamics with Euler integration — drive states are bilaterally coupled (synchronization), turn states have mutual inhibition — and (3) body feedback: velocity mismatch drives stance correction, body instability dampens rhythm, contact asymmetry produces corrective turning, and slip detection boosts stance. State variables are saturated to prevent runaway. Output mapping converts state to locomotion commands through tanh nonlinearities: $\text{forward_drive} = 0.1 + 0.9 \cdot \tanh(\text{mean_drive})$, $\text{turn_drive} = \tanh(\text{turn_L} - \text{turn_R})$, $\text{step_frequency} = 1.0 + 1.5 \cdot \tanh(\text{rhythm})$, $\text{stance_gain} = 1.0 + 0.5 \cdot \tanh(\text{stance})$.

4.7 Segregation analysis

For each sensory modality group (olfactory: 100 neurons; visual/LPLC2: 310; somatosensory: 75 across four subchannels), we identified all descending neurons receiving direct synaptic input (1-hop) from the full connectome. For 2-hop analysis, we identified all intermediate neurons that are both direct postsynaptic targets of sensory neurons and direct presynaptic to readout DNs. Overlap was quantified using the Jaccard index for all pairwise comparisons at both the DN level and the interneuron level. Excitatory/inhibitory classification used the Excitatory column from the connectome. Subchannel analysis repeated the 1-hop computation for each of the 10 individual sensory channels.

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